1. Abstract

TurfHub is a mobile-first web application designed to streamline the discovery, reservation, and management of artificial football turf facilities across Dhaka, Bangladesh. The platform tackles fragmented manual bookings, unverified peer-to-peer mobile payments, and double-booking conflicts by introducing an OpenStreetMap geospatial discovery engine and a 10-minute pessimistic slot-locking mechanism. It unifies players, turf operators, and platform administrators into a cohesive ecosystem backed by verified bKash and SSLCOMMERZ transactions. Ultimately, TurfHub establishes a modern, reliable digital infrastructure that modernizes recreational sports scheduling across Dhaka.

2. Problem Statement

Recreational football on artificial turf pitches is rapidly growing in Dhaka, but venue discovery and booking remain severely bottlenecked by fragmented social media announcements and manual phone calls. Advance deposits are transacted via unstructured personal mobile financial transfers without atomic verification, frequently causing double-booking race conditions during peak hours. Furthermore, severe urban traffic congestion makes proximity-based venue selection critical, yet players lack localized maps showing real-time slot availability, pitch dimensions, and parking. Simultaneously, venue owners lack digital tooling and rely on physical notebooks, resulting in untracked revenue, operational inefficiencies, and missed peak pricing opportunities.

3. Purpose of Project

The primary purpose of TurfHub is to replace Dhaka's disjointed sports booking workflows with a centralized, automated digital platform. It aims to guarantee zero booking overlaps through atomic multi-document transactions and an automated 10-minute pessimistic lock timer. The project delivers sub-250ms geospatial discovery across Dhaka's key zones while adhering to mobile-first WCAG AA accessibility standards.

4. Proposed Solution

TurfHub provides a unified three-tiered web platform tailored specifically for players, venue operators, and platform administrators. Players can locate nearby pitches via an interactive Leaflet/OpenStreetMap interface, place an atomic 10-minute hold on desired slots, and pay advance fees securely via verified bKash and SSLCOMMERZ gateways. Turf owners receive a dedicated operations portal to manage daily multi-pitch schedules, enforce peak versus off-peak rate tiers, block maintenance hours, and track real-time revenue. Platform administrators oversee ecosystem integrity through an inspection queue that verifies trade licenses, manages listings, and monitors live server logs.

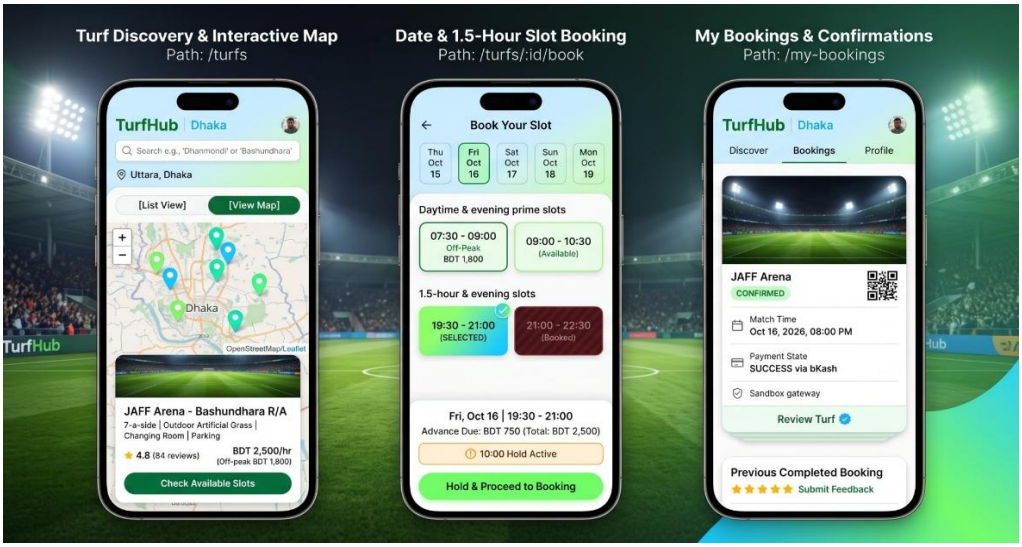
5. Technology Stack

- **Frontend:** React 18 (Vite), Tailwind CSS
- **Backend:** Node.js, Express.js (RESTful Architecture)
- **Database:** MongoDB Atlas (MERN Stack)

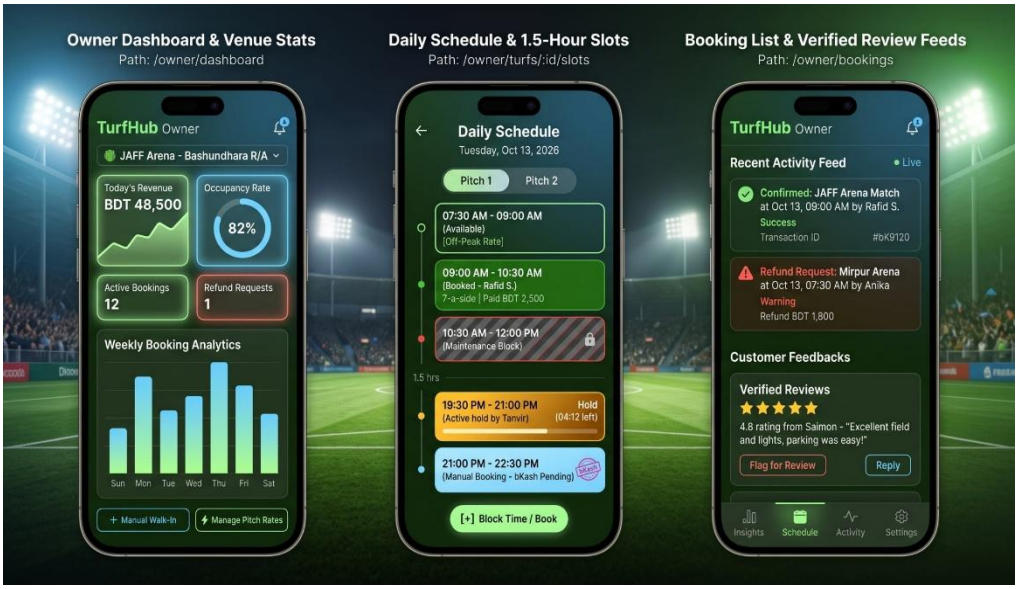
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6. UI Screenshots

Section 1: Customer Portal UI



Section 2: Owner Portal UI:



Section 3: Platform Admin Portal UI

